



# What Happens When an Automated System Makes the Wrong Decision — And There Is No Record?

OMNIX is the governance infrastructure that ensures every automated decision is evaluated, authorized, and cryptographically sealed — before it executes.

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**\$137B+**

Total Addressable Market

**11**

Governance Checkpoints per Decision

**24/7**

Production Operation — Live Now

Eureka GCC Dubai 2026  
Semifinalist

SSRN · 2 Papers 6321298 ·  
6507559

Zenodo · 2 DOIs 19056919 ·  
19375792

Pre-Seed Round \$500K at \$3M  
pre-money

# The Real Cost of One Unaudited Decision

Every day, automated systems execute thousands of decisions that carry direct financial, legal, and reputational exposure. The question is not whether something will go wrong — it is whether you will be able to prove what happened when it does.

|                                                                      |                                                                         |                                                                            |
|----------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <div>€35M</div> <div>Maximum fine under EU AI Act per incident</div> | <div>\$50M+</div> <div>Average regulatory fine for AML violations</div> | <div>78%</div> <div>Of AI incidents leave no recoverable audit trail</div> |
|----------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|

*"When regulators ask what the system decided, why it decided it, and who authorized it — most organizations cannot answer any of the three."*

| Scenario                                    | Without OMNIX                                                                   | With OMNIX                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Automated trade executes incorrectly        | No signed record. Dispute resolution takes months. Potential liability.         | Signed receipt proves what was evaluated and authorized. Dispute resolved in hours.    |
| Regulator requests AI decision audit        | Logs may be incomplete or missing. No cryptographic proof. Non-compliance risk. | Every decision carries a tamper-evident receipt. Full audit trail delivered instantly. |
| Insurance claim denied by automated system  | Client disputes denial. No verifiable record of evaluation criteria applied.    | Receipt shows all 11 checkpoints passed or which one blocked the decision.             |
| Autonomous system takes unauthorized action | No pre-execution gate. System acted without independent governance approval.    | Fail-closed pipeline blocked execution. Receipt proves governance was applied.         |

# What OMNIX Does

OMNIX sits between your automated system and the moment a decision executes. It evaluates every proposed decision through 11 independent governance checkpoints and issues a cryptographically-sealed receipt. No receipt — no execution.



*If any checkpoint fails → pipeline halts → decision is BLOCKED → signed receipt issued.*

| Decision Result | What It Means                                                                    | Receipt Issued?                            |
|-----------------|----------------------------------------------------------------------------------|--------------------------------------------|
| APPROVED        | All 11 checkpoints authorized the decision. Execution proceeds.                  | Yes — signed proof of authorization        |
| BLOCKED         | One or more checkpoints identified a governance violation. Execution halted.     | Yes — signed proof of rejection and reason |
| HOLD            | Internal signal contradiction detected. Human review required before proceeding. | Yes — signed escalation record             |

## The 11 Governance Checkpoints

Each decision submitted to OMNIX passes through 11 independent checkpoints in sequence. Every checkpoint operates autonomously — a failure at any stage halts the pipeline immediately and issues a signed rejection receipt. No checkpoint can be bypassed or overridden.

| CP    | Checkpoint                       | What It Evaluates                                                                                                                     | If It Fails                              |
|-------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| CP-1  | Signal Integrity Validator (SIV) | Verifies data quality, completeness, and input integrity before any evaluation begins.                                                | Rejected at entry. Pipeline never opens. |
| CP-2  | Probability Assessment           | Evaluates statistical confidence in the proposed decision outcome against authorized thresholds.                                      | Blocked. Insufficient confidence.        |
| CP-3  | Risk Evaluation                  | Quantifies downside exposure and compares against the authorized risk envelope for the domain.                                        | Blocked. Risk limit exceeded.            |
| CP-4  | Coherence Engine                 | Detects internal contradictions across all active signals. Decision Contradiction Index (DCI) $\geq 70$ mandates escalation.          | HOLD. Human review required.             |
| CP-5  | Trend Validator                  | Confirms alignment between the proposed decision and the prevailing operational or market regime.                                     | Blocked. Regime contradiction.           |
| CP-6  | Stress Testing                   | Simulates adverse conditions (liquidity shocks, volatility spikes) to validate decision robustness.                                   | Blocked. Fails under stress.             |
| CP-7  | Ethics & Domain Gate             | Enforces domain-specific ethical constraints: Sharia compliance (riba, gharar), safety limits for robotics, bias controls for credit. | Blocked. Ethics violation recorded.      |
| CP-8  | Threshold & Context Validator    | Validates all decision parameters against authorized operational boundaries and contextual constraints.                               | Blocked. Parameter out of range.         |
| CP-9  | AML Screening                    | Screens for anti-money laundering indicators, suspicious transaction patterns, and sanctioned entity exposure.                        | Blocked. Suspicious activity flagged.    |
| CP-10 | Fraud Detection                  | Multi-layer fraud signal analysis across behavioral, transactional, and systemic patterns.                                            | Blocked. Fraud flag escalated.           |
| CP-11 | Jurisdiction Compliance          | Verifies regulatory jurisdiction eligibility before any cross-border or regulated decision executes.                                  | Blocked. Jurisdiction violation.         |

*"Every checkpoint is independent. The system cannot approve a decision because 10 out of 11 passed. All 11 must pass. This is what fail-closed governance means."*

# The Business Case: ROI & Value Delivered

OMNIX is not a cost center — it is a risk-reduction investment with a quantifiable return. The value delivered falls into three categories: regulatory protection, operational efficiency, and reputational capital.

|                                                                            |                                                                    |                                                                       |
|----------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|
| <div>\$500K+</div> <div>Minimum regulatory fine avoided per incident</div> | <div>100%</div> <div>Audit trail coverage for every decision</div> | <div>&lt;1ms</div> <div>Governance evaluation time per decision</div> |
|----------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|

| Value Category       | Without OMNIX                              | With OMNIX                               | Estimated Impact                        |
|----------------------|--------------------------------------------|------------------------------------------|-----------------------------------------|
| Regulatory Fines     | No audit trail. Full exposure.             | Signed receipt per decision.             | €35M EU AI Act exposure eliminated      |
| Incident Response    | Weeks of log reconstruction.               | Instant receipt retrieval.               | 80% reduction in response time          |
| Legal Disputes       | He said / she said. No proof.              | Cryptographic proof of what was decided. | Disputes resolved in hours, not months  |
| Compliance Reporting | Manual, error-prone, expensive to produce. | Automated, signed, instantly accessible. | 60–80% reduction in compliance cost     |
| Client Trust         | Black box — cannot explain decisions.      | Public verification URL per decision.    | Measurable increase in client retention |

## Simple ROI Calculation

OMNIX Enterprise cost: \$20,000 – \$35,000 / month

One avoided EU AI Act fine: €35,000,000

Break-even point: 1 avoided incident = 145 years of OMNIX cost

The question is not whether OMNIX is expensive.

The question is whether you can afford to operate without it.

# Four Active Governance Domains

The same governance pipeline operates across four distinct industries simultaneously. OMNIX does not need to be rebuilt for each domain — it is configured. This is what makes it infrastructure, not software.

| Domain                    | What OMNIX Governs                                                    | Key Risk Eliminated                             | Status      |
|---------------------------|-----------------------------------------------------------------------|-------------------------------------------------|-------------|
| Digital Asset Trading     | Every trade entry and exit decision. Real capital on Kraken exchange. | Unauthorized execution under adverse conditions | LIVE 24/7   |
| Islamic Credit GCC / MENA | Murabaha financing decisions. Sharia compliance enforced at CP-6.     | Riba violation and non-compliant structuring    | OPERATIONAL |
| Insurance Claims          | Automated claim assessment and fraud screening decisions.             | Fraudulent approval and unaudited denial        | OPERATIONAL |
| Robotics Pre-Execution    | Autonomous action authorization before any physical command executes. | Unauthorized physical action — safety breach    | OPERATIONAL |

*"The same 11-checkpoint pipeline that governs a crypto trade also governs a Murabaha credit decision. Domain changes. Governance does not."*

# The Governance Receipt: Your Proof of Accountability

Every decision processed by OMNIX generates a receipt. This receipt is the evidence that governance was applied. It is cryptographically signed, tamper-evident, and independently verifiable by anyone — regulators, auditors, counterparties, or clients.

```
{
  "receipt_id": "omnix-gov-20260401-a7f3c9",
  "timestamp": "2026-04-01T14:32:17.841Z",
  "decision": "APPROVED",
  "checkpoints_passed": 11,
  "checkpoints_total": 11,
  "veto_count": 0,
  "governance_summary": "All checkpoints passed",
  "cryptographic_seal": {
    "algorithm": "PQC signature (OMNIX implementation)",
    "merkle_root": "a3f7c2d8e1b490fa231c...",
    "verification_url": "https://omnixquantum.net/verify/..."
  }
}
```

| Field              | What It Proves                                                                      |
|--------------------|-------------------------------------------------------------------------------------|
| receipt_id         | Unique link to the complete decision audit record                                   |
| decision           | The governance determination: APPROVED, BLOCKED, or HOLD                            |
| checkpoints_passed | How many of the 11 checkpoints independently authorized the decision                |
| cryptographic_seal | Post-quantum signature — mathematically impossible to forge or alter after issuance |
| verification_url   | Public link — any third party can independently confirm this receipt at any time    |

# Regulatory Protection Built In

Governance receipts are the evidence layer that regulators are beginning to require. OMNIX was designed with these frameworks as first-class requirements — not added as an afterthought.

| Framework        | Requirement                                                                       | OMNIX Coverage                                                                            |
|------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| EU AI Act        | High-risk AI systems must maintain decision logs and enable human oversight.      | Every decision: immutable signed receipt + HOLD escalation for human review               |
| NIST AI RMF      | AI systems must be measurable, explainable, and subject to governance controls.   | 11 checkpoints provide structured, measurable evaluation. Receipts enable explainability. |
| ISO/IEC 42001    | AI management systems require documented risk controls and incident traceability. | Fail-closed pipeline + public verification URL = full incident traceability               |
| AAOIFI (Islamic) | Sharia-compliant products must adhere to riba and gharar prohibitions.            | CP-6 enforces Sharia principles as a pre-execution governance gate                        |



## Why Not Just Use What You Already Have?

Every organization already has some form of logging, monitoring, or compliance tooling. The question is not whether those tools exist — it is whether they provide pre-execution governance with cryptographic accountability. They do not.

| Alternative                | What It Does                                                        | What's Missing                                                                                 | OMNIX Difference                                                                                      |
|----------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Audit Logs                 | Record what happened after execution has already occurred.          | Evidence created after the fact. Cannot prevent a bad decision.                                | OMNIX governs BEFORE execution. The receipt proves authorization, not reconstruction.                 |
| SIEM / Monitoring          | Alert teams to anomalies once they are detected.                    | Cannot halt execution. Cannot produce signed proof of evaluation. Observation only.            | OMNIX halts. Fail-closed by design. No signed receipt = no execution. Ever.                           |
| Compliance Platforms       | Report policy adherence and flag violations after the fact.         | Policy check ≠ governance gate. No cryptographic seal. Cannot be independently verified.       | OMNIX enforces. Every decision carries a tamper-evident, publicly verifiable signed receipt.          |
| Custom Internal Validation | Domain-specific code checks written by internal engineering teams.  | Non-standardized. No independent verification. Cannot audit externally. Breaks across domains. | OMNIX is domain-agnostic infrastructure. Same pipeline, same receipt standard, across every vertical. |
| Human Review Committees    | Manual oversight for high-stakes decisions by designated reviewers. | Does not scale. Inconsistent. No cryptographic record of what was reviewed and why.            | OMNIX escalates to human review only when the Coherence Engine detects contradiction (CP-4 HOLD).     |

*"The difference between an audit trail and governance infrastructure is the same as the difference between a black box recorder and a pilot. One tells you what happened. The other determines whether it should happen at all."*

## How to Start

OMNIX is designed to integrate without disrupting your existing systems. The Shadow Mode tier allows full governance evaluation in parallel with your current operations — zero risk, immediate visibility.

| Tier        | How It Works                                                                                   | Best For                                                                                | Investment                  |
|-------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------|
| Shadow Mode | OMNIX runs in parallel. Evaluates every decision. No intervention. Full receipt generation.    | Organizations that want visibility before committing to full governance.                | Complimentary               |
| Advisory    | OMNIX results delivered as recommendations. Your team retains final decision authority.        | Regulated environments where human oversight is currently mandatory.                    | \$8,000 / month             |
| Enterprise  | Full integration. OMNIX receipt required before any automated decision executes. Fail-closed.  | Organizations with significant regulatory exposure or high-stakes automated operations. | \$20,000 – \$35,000 / month |
| Custom      | Multi-domain deployment, private infrastructure, custom governance parameters, SLA guarantees. | Financial institutions, government, and infrastructure operators at scale.              | Contact us                  |

# Operational Track Record

OMNIX has been in continuous production operation since July 2025, with the governance pipeline actively evaluating decisions across all four domains. The system has maintained 24/7 uptime with zero infrastructure failures.

*Note: Digital asset trading operates with real capital on Kraken exchange. Islamic credit, insurance, and robotics domains operate in a controlled validation environment — governance logic is identical; domain-specific parameters are under calibration for production deployment.*

| Milestone               | Date          | Significance                                                                                                                                                                  |
|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Production deployment   | July 2025     | Live operation with real capital on Kraken exchange. Governance pipeline validated under live market conditions.                                                              |
| PQC implementation      | November 2025 | Post-quantum cryptographic signature deployed. All receipts carry NIST-standardized PQC signatures.                                                                           |
| Multi-domain expansion  | March 2026    | Islamic credit, insurance, and robotics domains activated in validation environment. Same pipeline across all four sectors.                                                   |
| Track Record phase      | January 2026  | Recalibrated pipeline activated. Official governance performance measurement period begins.                                                                                   |
| Academic publications   | March 2026    | 4 peer-indexed publications: SSRN abstract_id=6321298 (PQC architecture) and 6507559 (governance whitepaper); Zenodo DOI 10.5281/zenodo.19056919 and 10.5281/zenodo.19375792. |
| Eureka GCC Semifinalist | 2026          | Selected as Semifinalist for Eureka GCC Dubai 2026 — competitive pan-GCC innovation program.                                                                                  |

# Why Now: The Regulatory Window Is Closing

For the first time, AI governance is not a recommendation — it is a legal mandate. Organizations that acquire governance infrastructure now will be compliant by default. Those that wait will face retrofit costs that dwarf the investment.

| Framework / Event           | Deadline / Status     | Implication for Automated Systems                                                                                           |
|-----------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| EU AI Act — High-Risk AI    | August 2026           | Mandatory audit trails, human oversight mechanisms, and explainability for all high-risk AI systems.                        |
| NIST AI RMF                 | Adopted — 2023        | De facto standard for US federal agencies and regulated industries. Measurability and governance controls required.         |
| UAE AI Governance Framework | Active — 2025         | UAE mandates AI accountability standards across financial services, healthcare, and public sector.                          |
| SEC / FCA / MAS Guidance    | Published — 2024–2025 | Financial regulators across US, UK, and Singapore have published AI governance expectations for trading and credit systems. |
| ISO/IEC 42001               | Published — 2023      | International AI management system standard. Requires documented risk controls and incident traceability.                   |

## The Founder

### Harold Nunes

Solo Founder & CEO — OMNIX

While building automated trading systems, Harold identified a gap that no existing product addressed: the absence of pre-execution governance with cryptographic accountability across any domain.

Rather than wait for someone else to build it, he built it himself — from architecture to production deployment, without a corporate engineering team.

**What he built:** An 11-checkpoint governance pipeline, post-quantum cryptographic signatures (NIST-standardized), multi-domain deployment across trading, Islamic credit, insurance, and robotics — and 4 peer-indexed academic publications documenting the architecture.

"I couldn't find infrastructure that governed automated decisions with cryptographic accountability. So I built it."

Currently seeking: \$500K pre-seed · Strategic co-founder · First enterprise clients

## Contact & Next Steps

**Harold Nunes**

Founder & CEO — OMNIX

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**Three ways to move forward:**

1. **Start with Shadow Mode** — free, no disruption, immediate governance visibility.
2. **Schedule a technical demo** — see the pipeline evaluate a real decision live.
3. **Request the full technical specification** — available under NDA for institutional due diligence.

**OMNIX is raising \$500,000 USD pre-seed at a \$3,000,000 pre-money valuation. For investor inquiries contact Harold Nunes directly.**

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